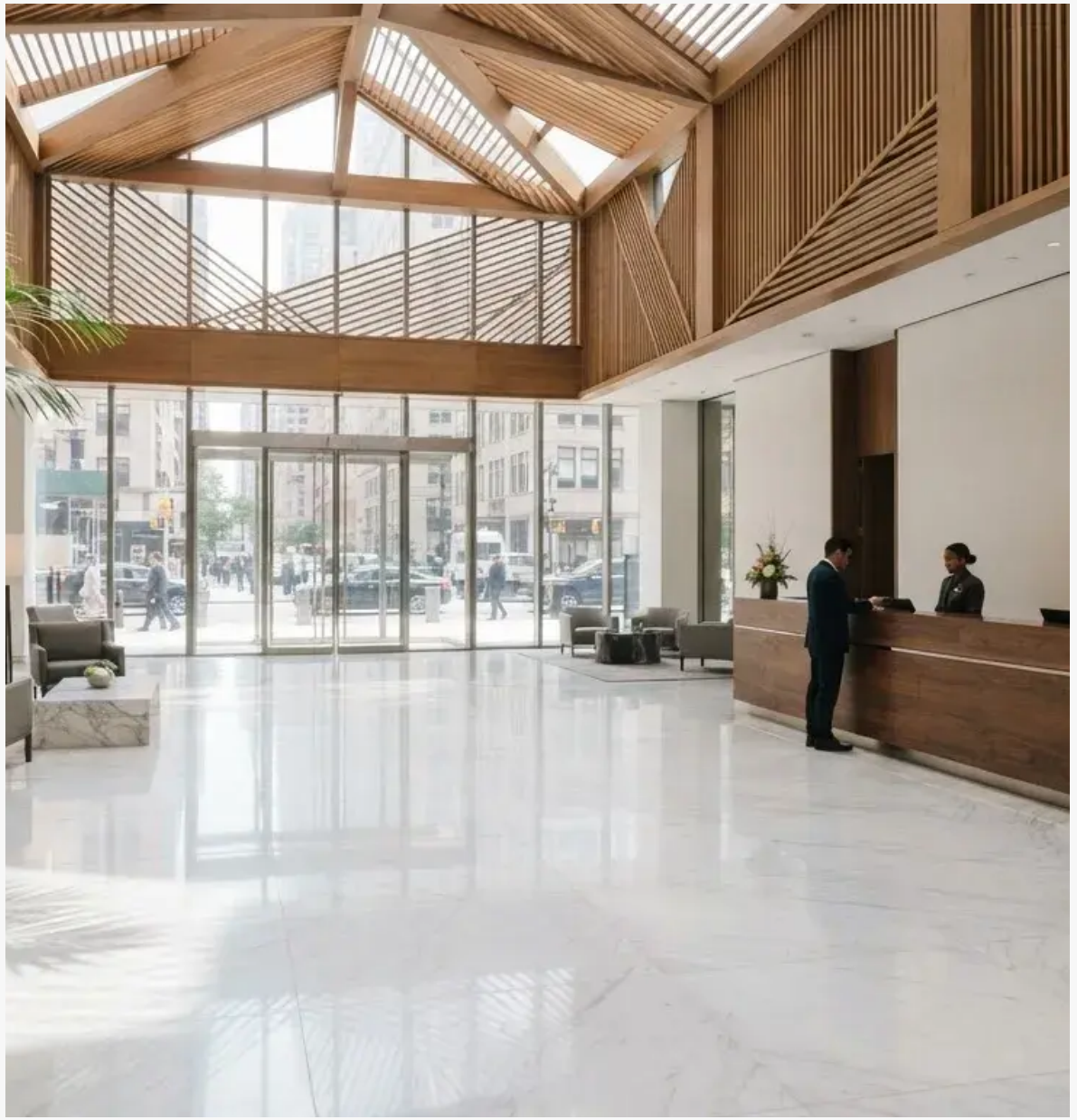
Database Systems

Hotel Reservation & Guest Services DBMS

Design and Implementation Overview



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System Requirements & Entities

Identifying the core components of the hotel management ecosystem

Core Entities

Guests, Rooms, Reservations, and Stays forming the backbone of the system

Service Management

Housekeeping tasks, service requests, and associated service charges

Financials

Rate plans, invoices, and payment processing for accurate billing

User Roles

Access control for Admin, Receptionist, Housekeeping, and Guests

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Database Design & Normalization

Ensuring data integrity through structured relational schema



ER Modeling

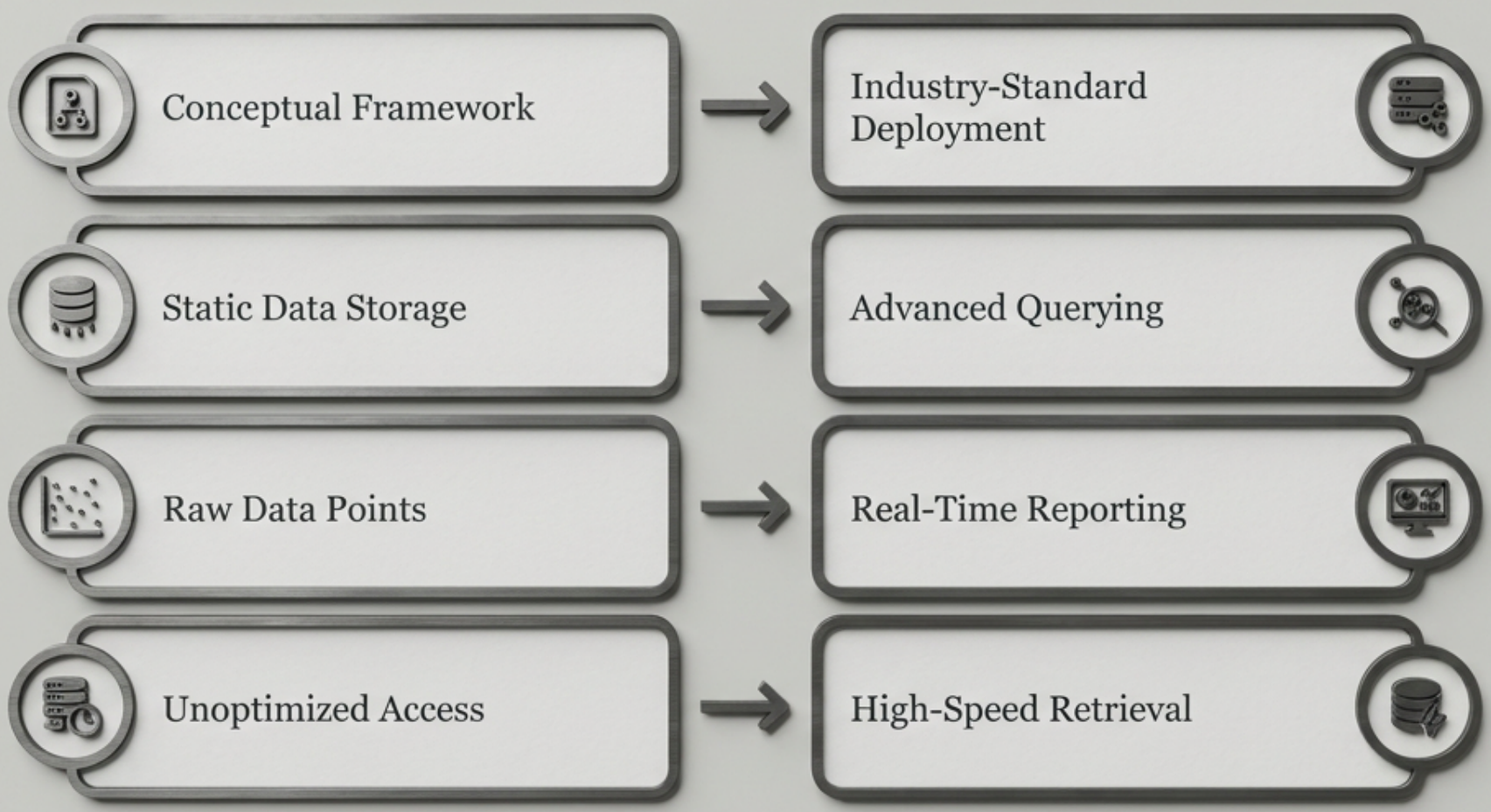
Normalization

Integrity Constraints

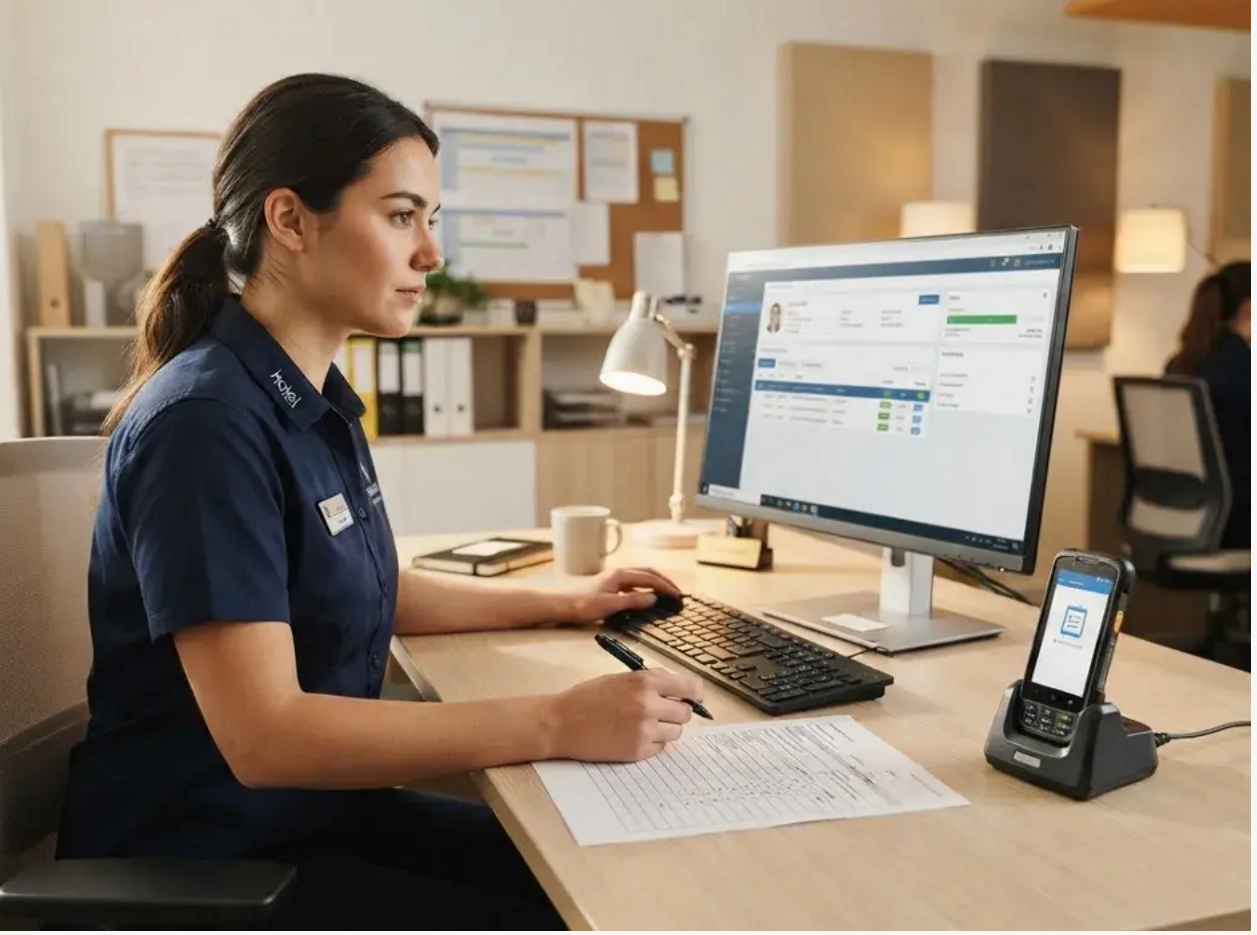
Business Rules

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Implementation & SQL Reporting



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Front-End & System Testing

The user interface and validation layer for hotel operations

Application Layer

Java or Python front-end supporting availability search and billing.

Core Features

Workflows for check-in/out, service posting, and housekeeping assignment.

Validation

Input validation and transaction awareness to maintain data consistency.

Testing

Systematic verification of reservation flows and payment limit enforcement.

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